SE LAB 5

Static Code Analysis (Pylint, Bandit, Flake8)

Name: Inchara K Kuppal | SRN: PES1UG25CS818

Assigned script: election_system.py (L5_62_voting-election-system)

1. Before and After Summary

Tool	Before	After
Pylint score	5.33 / 10	10.00 / 10
Bandit issues	7 (2 High, 1 Medium, 4 Low)	0
Flake8 issues	30	0

2. Issues Found and Fixed

Line numbers refer to the original script.

Issue	Type	Line(s)	Description	Fix Approach
Weak MD5 hash (B324)	Security (High)	17	hashlib.md5 is a broken hash and was used on voter IDs.	Switched to hashlib.sha256.
Hidden bidi character (B613, E2502)	Security / Bug (High)	22	A hidden \u202e character pulled the comment into the string, so isOfficer() gave the wrong answer.	Rewrote as return role != "voter" with a clean comment.
Unsafe XML parsing (B409, B319)	Security (Medium)	7, 75	xml.dom.pulldom is open to XML attacks on untrusted input.	Used defusedxml.pulldom.
Hardcoded password (B105)	Security (Low)	9	Admin password was written in the source code.	Read it from the ADMIN_PASSWORD environment variable.
Insecure random (B311)	Security (Low)	72	random is predictable, not good for receipt numbers.	Used secrets.randbelow().
try / except / pass, bare except (B110, W0702, E722)	Security / Quality	82-83	Errors were silently ignored.	Catch AttributeError and log a warning.
Lambda in loop (W0640)	Bug	68	All lambdas used the last name, so NOTA: 0 printed three times.	Replaced with a list comprehension; output is now correct.
Unused imports (F401, W0611)	Quality	1-3	os, sys, json were never used.	Removed sys and json; os now used for the password.
Unused variables (F841, W0612)	Quality	32, 45, 55	receipt, total and now were assigned but never used.	Removed them; receipt is now stored on the voter.
camelCase function names (C0103)	Style	15, 21, 26, 51, 58, 71, 74	Names did not follow snake_case.	Renamed, e.g. registerVoter to register_voter.
Missing docstrings (C0114, C0116)	Style	whole file	No module or function docstrings.	Added short docstrings.
Shadowed global name (W0621)	Quality	58	Parameter ballots hid the global list ballots.	Renamed it to ballot_list.
Blank lines, trailing spaces, long lines (E302, E305, W291, E501, C0303, C0301)	Format	many	PEP 8 formatting problems.	Fixed spacing, split long lines, added a poll_time() helper.
Missing final newline (C0304)	Style	end of file	File did not end with a newline.	Added the newline.

3. Commits

1. Fix Bandit issues: MD5, bidi char, XML parsing, hardcoded password, random, try-except-pass
2. Fix Pylint issues: naming, docstrings, unused code, closure bug, whitespace
3. Fix Flake8 issues: blank lines, line length, final newline
Before and after reports for all three tools are in the repo.

4. Repository

https://github.com/inchara13kuppal/PES1UG25CS818_SE_Assignments (folder: Lab5)